II. THE KOGUT-SUSSKIND HAMILTONIAN FOR QCD AND MAPPING TO QUBITS

The 1+1D SU(3) KS Hamiltonian [89, 90] with N_f flavors formulated in $A_x^{(a)} = 0$ gauge [37, 91] takes the form

$$H = \sum_f \left[\frac{1}{2} \sum_{n=0}^{2L-2} \left(\phi_n^{(f)\dagger} \phi_{n+1}^{(f)} \; + \; \text{h.c.} \right) \; + \; m_f \sum_{n=0}^{2L-1} (-1)^n \phi_n^{(f)\dagger} \phi_n^{(f)} \right] \; + \; \frac{g^2}{2} \sum_{n=0}^{2L-2} \sum_{a=1}^{8} \left(\sum_{m \leq n} Q_m^{(a)} \right)^2 \; , \tag{1}$$

where $\phi_n^{(f)}$ correspond to annihilation operators for fermions of flavor f. They are color triplets, with their color indices suppressed in Eq. (1). The color-charge operators on each lattice site are the sum of contributions from each flavor. For example, for $N_f = 2$ (up and down quarks), the color-charge operators are

$$Q_m^{(a)} \; = \; \phi_m^{(u)\dagger} T^a \phi_m^{(u)} \; + \; \phi_m^{(d)\dagger} T^a \phi_m^{(d)} \; , \tag{2}$$

where the generators of SU(3), T^a, are given in App. A. With open boundary conditions (OBC) and vanishing fields at spatial infinity, corresponding to vanishing net color charge on the lattice (enforced by additional terms in the Hamiltonian [37]), Gauss's law is sufficient to determine the chromo-electric field at all lattice sites,

$$\mathbf{E}_n^{(a)} = \sum_{m \leq n} Q_m^{(a)} \; . \tag{3}$$

There are a number of ways that this system, with the Hamiltonian given in Eq. (1), can be mapped onto qubit registers. In our previous works [37, 39], the KS Hamiltonian for an arbitrary number of colors N_c and flavors N_f was mapped to qubits using the Jordan-Wigner (JW) transformation [93]. For the $N_c = 3$ and $N_f = 2$ case, each staggered site requires six qubits, with ordering $d_b, d_g, d_r, u_b, u_g, u_r$, and the antiquarks associated with the same spatial site adjacent with ordering $\overline{d}_b, \overline{d}_g, \overline{d}_r, \overline{u}_b, \overline{u}_g, \overline{u}_r$. This is shown in the left panel of Fig. 1. The resulting JW-mapped

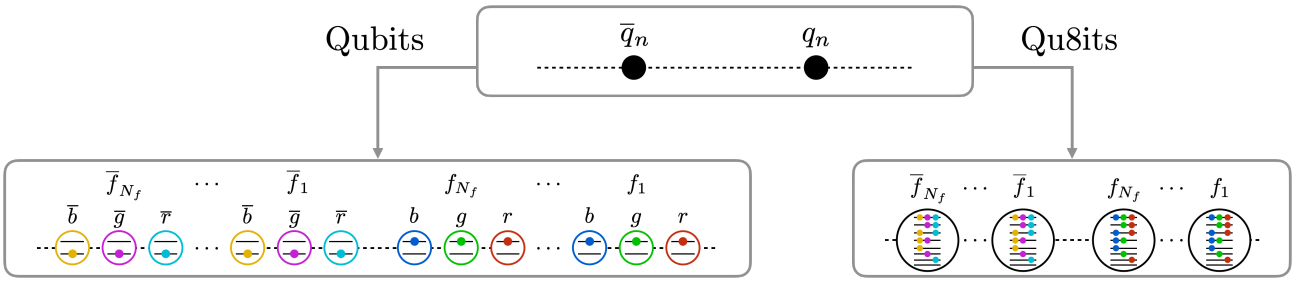

FIG. 1. Mapping QCD with N_f quark flavors onto a lattice of qubits (left) or qu8its (right) describing a spatial site. Kogut-Susskind (staggered) fermions are used for the quark fields, with (anti)quarks on (odd) even sites. Using qubits, color and flavor degrees of freedom of each quark and antiquark site are distributed over six qubits with a JW mapping. Using qu8its, with the quark (and anti-quark) degrees of freedom being mapped to the internal states, only two qu8its are required per each quark flavor.

Hamiltonian is the sum of three terms [37, 39], neglecting the possible presence of chemical potentials,

$$\begin{aligned}
H \; &= \; H_{kin} \; + \; H_m \; + \; H_{el} \; , \\
H_{kin} \; &= \; -\frac{1}{2} \sum_{n=0}^{2L-2} \sum_{f=0}^{1} \sum_{c=0}^{2} \left[\sigma^+_{6n+3f+c} \left(\bigotimes_{i=1}^{5} \sigma^z_{6n+3f+c+i} \right) \sigma^-_{6(n+1)+3f+c} + \text{h.c.} \right] \; , \\
H_m \; &= \; \frac{1}{2} \sum_{n=0}^{2L-1} \sum_{f=0}^{1} \sum_{c=0}^{2} m_f \left[(-1)^n \sigma^z_{6n+3f+c} + 1 \right] \; , \\
H_{el} \; &= \; \frac{g^2}{2} \sum_{n=0}^{2L-2} (2L-1-n) \left(\sum_{f=0}^{1} Q_{n,f}^{(a)} Q_{n,f}^{(a)} + 2 Q_{n,0}^{(a)} Q_{n,1}^{(a)} \right) + g^2 \sum_{n=0}^{2L-3} \sum_{m=n+1}^{2L-2} (2L-1-m) \sum_{f=0}^{1} \sum_{f'=0}^{1} Q_{n,f}^{(a)} Q_{m,f'}^{(a)} \; ,
\end{aligned} \tag{4}$$

where repeated adjoint color indices (a) are summed over, the flavor indices $f = \{0, 1\}$ correspond to u- and d-quark

flavors, and $\sigma^{\pm} = (\sigma^x \pm i\sigma^y)/2$. Products of charges, given in terms of spin operators, are given in Refs. [37, 39],

$$
\begin{aligned}
Q^{(a)}_{n,f}\, Q^{(a)}_{n,f} &= \frac{1}{3}(3 - \sigma^z_{6n+3f}\sigma^z_{6n+3f+1} - \sigma^z_{6n+3f}\sigma^z_{6n+3f+2} - \sigma^z_{6n+3f+1}\sigma^z_{6n+3f+2}) \ , \\
Q^{(a)}_{n,f}\, Q^{(a)}_{m,f'} &= \frac{1}{4}\Big[2\big(\sigma^+_{6n+3f}\sigma^-_{6n+3f+1}\sigma^-_{6m+3f'}\sigma^+_{6m+3f'+1} + \sigma^+_{6n+3f}\sigma^z_{6n+3f+1}\sigma^-_{6n+3f+2}\sigma^-_{6m+3f'}\sigma^z_{6m+3f'+1}\sigma^+_{6m+3f'+2} \\
&\quad + \sigma^+_{6n+3f+1}\sigma^-_{6n+3f+2}\sigma^-_{6m+3f'+1}\sigma^+_{6m+3f'+2} + \text{h.c.}\big) + \frac{1}{6}\sum_{c=0}^{2}\sum_{c'=0}^{2}(3\delta_{cc'} - 1)\sigma^z_{6n+3f+c}\sigma^z_{6m+3f'+c'}\Big] \ . \qquad (5)
\end{aligned}
$$

A constant has been added to H_m so that all basis states contribute a positive mass.

The time-evolution operator corresponding to this Hamiltonian can be implemented by sequences of unitary operators. These Trotterized quantum circuits, and hence the associated gate counts per Trotter step, have been determined in Refs. [37, 39].

III. QU8ITS FOR QCD

The main drivers for considering mapping the KS Hamiltonian to qudits with $d = 8$, as discussed above, is to reduce the number of two-qudit entangling gates from the number required for qubits. The single-component fermion formulation (per color state) that defines the staggering of the quark fields draws comparisons with the constructions used in describing nuclear many-body systems. We recently showed the utility of using $d = 5$ qudits (qu5its) to describe spin-paired multi-nucleon systems in the context of the Agassi model [66]. We make use of this analogy in mapping the quark (and anti-quark) fields to qu8its, as displayed in the right panel of Fig. 1.

A. Mapping Quarks and Anti-Quarks to Qu8its

For a single flavor quark staggered site and using the JW mapping to three qubits, the qubits define the occupation of the three colors q_r, q_g, q_b throughout a quantum simulation. Satisfying fermion anti-commutation relations, the mapping of color occupations to qu8its can be chosen to be,

$$
\begin{aligned}
\big\{|\text{qu8it}\rangle\big\} &= \big\{|\Omega\rangle \ , \ |q_r\rangle \ , \ |q_g\rangle \ , \ |q_b\rangle \ , \ |q_g q_b\rangle \ , \ -|q_r q_b\rangle \ , \ |q_r q_g\rangle \ , \ |q_r q_g q_b\rangle\big\} \\
&= \big\{|\Omega\rangle \ , \ \hat{c}^\dagger_r|\Omega\rangle \ , \ \hat{c}^\dagger_g|\Omega\rangle \ , \ \hat{c}^\dagger_b|\Omega\rangle \ , \ \hat{c}^\dagger_g\hat{c}^\dagger_b|\Omega\rangle \ , \ -\hat{c}^\dagger_r\hat{c}^\dagger_b|\Omega\rangle \ , \ \hat{c}^\dagger_r\hat{c}^\dagger_g|\Omega\rangle \ , \ \hat{c}^\dagger_r\hat{c}^\dagger_g\hat{c}^\dagger_b|\Omega\rangle\big\} \\
&= \big\{|1\rangle \ , \ |2\rangle \ , \ |3\rangle \ , \ |4\rangle \ , \ |5\rangle \ , \ |6\rangle \ , \ |7\rangle \ , \ |8\rangle\big\} \ , \qquad (6)
\end{aligned}
$$

where the fermionic vacuum state is $|\Omega\rangle$ and the $\hat{c}_\alpha$ operators are elements of the $\phi^{(f)}_j$ defined below Eq. (1).[6] At first, the group structure might be a little confusing, for instance the reason as to why there is only one state associated with the three quarks in the maximally occupied state ($|8\rangle$). Given that quarks reside in the fundamental representation of SU(3), $\mathbf{3}$, products of two and three quarks give rise to the following irreducible representations (irreps): $\mathbf{3}\otimes\mathbf{3} = \mathbf{6}\oplus\overline{\mathbf{3}}$ and $\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3} = \mathbf{10}\oplus\mathbf{8}\oplus\mathbf{8}\oplus\mathbf{1}$. For spinless fermions at the same lattice site, only the total antisymmetric irreps are allowed; therefore, the irreps of the mapping in Eq. (6) are constrained to be $\{\mathbf{1},\mathbf{3},\mathbf{3},\mathbf{3},\overline{\mathbf{3}},\overline{\mathbf{3}},\overline{\mathbf{3}},\mathbf{1}\}$, respectively. The symmetric representation $\mathbf{6}$, for example, is forbidden by antisymmetry. The states in Eq. (6) map naturally to the eight states of a single qu8it. Further details about this embedding can be found in App. B.

The anti-quarks are mapped to qu8its in an analogous way to the quarks, but with the replacement $\{r, g, b\} \to \{\overline{r}, \overline{g}, \overline{b}\}$,

$$
\begin{aligned}
\big\{|\overline{\text{qu8it}}\rangle\big\} &= \big\{|\Omega\rangle \ , \ |\overline{q}_r\rangle \ , \ |\overline{q}_g\rangle \ , \ |\overline{q}_b\rangle \ , \ |\overline{q}_g\overline{q}_b\rangle \ , \ -|\overline{q}_r\overline{q}_b\rangle \ , \ |\overline{q}_r\overline{q}_g\rangle \ , \ |\overline{q}_r\overline{q}_g\overline{q}_b\rangle\big\} \\
&= \big\{|\Omega\rangle \ , \ \hat{c}^\dagger_{\overline{r}}|\Omega\rangle \ , \ \hat{c}^\dagger_{\overline{g}}|\Omega\rangle \ , \ \hat{c}^\dagger_{\overline{b}}|\Omega\rangle \ , \ \hat{c}^\dagger_{\overline{g}}\hat{c}^\dagger_{\overline{b}}|\Omega\rangle \ , \ -\hat{c}^\dagger_{\overline{r}}\hat{c}^\dagger_{\overline{b}}|\Omega\rangle \ , \ \hat{c}^\dagger_{\overline{r}}\hat{c}^\dagger_{\overline{g}}|\Omega\rangle \ , \ \hat{c}^\dagger_{\overline{r}}\hat{c}^\dagger_{\overline{g}}\hat{c}^\dagger_{\overline{b}}|\Omega\rangle\big\} \\
&= \big\{|\overline{1}\rangle \ , \ |\overline{2}\rangle \ , \ |\overline{3}\rangle \ , \ |\overline{4}\rangle \ , \ |\overline{5}\rangle \ , \ |\overline{6}\rangle \ , \ |\overline{7}\rangle \ , \ |\overline{8}\rangle\big\} \ . \qquad (7)
\end{aligned}
$$

As is the case for the quark mapping, each state in Eq. (7) transforms as a single SU(3) irrep, which for the anti-quarks are $\{\mathbf{1},\overline{\mathbf{3}},\overline{\mathbf{3}},\overline{\mathbf{3}},\mathbf{3},\mathbf{3},\mathbf{3},\mathbf{1}\}$, respectively.

[6] The sign in the definition of $|6\rangle$ results from the ϵ^{132} in constructing a $\overline{\mathbf{3}}$ from $\mathbf{3}\otimes\mathbf{3}$.

With the mappings defined in Eqs. (6) and (7), the relevant operators for quantum simulation can be formed. The color-charge operator acting on the quarks is

$$\hat{Q}^{(a)} \;=\; \hat{\mathbf{c}}^\dagger T^a \hat{\mathbf{c}} \;,\;\; \hat{\mathbf{c}} \;=\; (\hat{c}_r, \hat{c}_g, \hat{c}_b)^T \;, \tag{8}$$

As the $\overline{\mathbf{3}}$ of di-quarks is also present in the mapping to qu8its, the color-charge operators acting on the anti-fundamental representation (the same as the anti-quarks) is also required,

$$\hat{\bar{Q}}^{(a)} \;=\; \hat{\bar{\mathbf{c}}}^\dagger \bar{T}^a \hat{\bar{\mathbf{c}}} \;,\;\; \hat{\bar{\mathbf{c}}} \;=\; \left(\hat{c}_{\bar{r}}, \hat{c}_{\bar{g}}, \hat{c}_{\bar{b}}\right)^T \;,\;\; \bar{T}^a \;=\; (-T^a)^* \;. \tag{9}$$

Therefore, the (8×8) matrix representations of the color-charge operator acting on qu8its and anti-qu8its are

$$\hat{Q}^{(a)} \;\to\; \widetilde{Q}^{(a)} \;=\; \left(\begin{array}{c|c|c|c} 0 & 0 & 0 & 0 \\ \hline 0 & T^a & 0 & 0 \\ \hline 0 & 0 & \bar{T}^a & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \;,\;\; \hat{\bar{Q}}^{(a)} \;\to\; \widetilde{\bar{Q}}^{(a)} \;=\; \left(\begin{array}{c|c|c|c} 0 & 0 & 0 & 0 \\ \hline 0 & \bar{T}^a & 0 & 0 \\ \hline 0 & 0 & T^a & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right) \;, \tag{10}$$

respectively, where the blocks of $\widetilde{Q}^{(a)}$ in Eq. (10) correspond to the action on $\{\mathbf{1}, \mathbf{3}, \overline{\mathbf{3}}, \mathbf{1}\}$ irreps, and the blocks of $\widetilde{\bar{Q}}^{(a)}$ correspond to the action on $\{\mathbf{1}, \overline{\mathbf{3}}, \mathbf{3}, \mathbf{1}\}$ irreps. A qu8it prepared in an arbitrary state $|\psi\rangle$, acted on by the color-charge operator, becomes

$$|\psi\rangle \;=\; \big\{|\text{qu8it}\rangle\big\} \cdot \boldsymbol{\xi} \;=\; \sum_{k=1}^{8} |k\rangle\, \xi_k \;,\;\; \hat{Q}^{(a)}|\psi\rangle \;=\; \big\{|\text{qu8it}\rangle\big\} \cdot \widetilde{Q}^{(a)} \cdot \boldsymbol{\xi} \;=\; \sum_{k=1}^{8} |k\rangle\, \left(\widetilde{Q}^{(a)} \cdot \boldsymbol{\xi}\right)_k \;, \tag{11}$$

where $\boldsymbol{\xi}$ is the vector of complex numbers defining the state. The baryon-number operator has a diagonal matrix representation,

$$\hat{\mathcal{B}} \to \widetilde{\mathcal{B}} \;=\; \frac{1}{3}\text{diag}\,(0, 1, 1, 1, 2, 2, 2, 3) \;. \tag{12}$$

Finally, the annihilation and creation operators acting on a qu8it have matrix representations

$$\widetilde{c_r} \;=\; \begin{pmatrix} 0&1&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&-1&0&0 \\ 0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \end{pmatrix} \;,\;\; \widetilde{c_g} \;=\; \begin{pmatrix} 0&0&1&0&0&0&0&0 \\ 0&0&0&0&0&0&-1&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \end{pmatrix} \;,\;\; \widetilde{c_b} \;=\; \begin{pmatrix} 0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&1&0&0 \\ 0&0&0&0&-1&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&0&0 \end{pmatrix} \;, \tag{13}$$

respectively, with $\widetilde{c_r}^\dagger = \widetilde{c_r}^T$, $\widetilde{c_g}^\dagger = \widetilde{c_g}^T$, and $\widetilde{c_b}^\dagger = \widetilde{c_b}^T$, satisfying the (required) fermionic anti-commutation relations, $\{\widetilde{c_\alpha}, \widetilde{c_\beta}^\dagger\} = \delta_{\alpha\beta}$ and $\{\widetilde{c_\alpha}, \widetilde{c_\beta}\} = \{\widetilde{c_\alpha}^\dagger, \widetilde{c_\beta}^\dagger\} = 0$ (with α, $\beta \in \{r, g, b\}$). The annihilation and creation operators act on the states analogously to the charge operators. For example, the action of $\hat{c}_r$ on a state $|\psi\rangle$ is

$$|\psi\rangle \;=\; \big\{|\text{qu8it}\rangle\big\} \cdot \boldsymbol{\xi} \;,\;\; \hat{c}_r|\psi\rangle \;=\; \big\{|\text{qu8it}\rangle\big\} \cdot \widetilde{c_r} \cdot \boldsymbol{\xi} \;. \tag{14}$$

The actions of the creation operators in Eq. (13) are shown in Fig. 2. The creation and annihilation operators acting on the anti-qu8it have the same representations as those acting on the qu8it in Eq. (13): $\widetilde{\bar{c}_\alpha} = \widetilde{c_\alpha}$.